Comparative Analysis of D Flip-Flop Using Static MOSFET and Transmission Gate in GPDK 180 nm and 90 nm Technologies

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***Abstract*—This paper presents the design and implementation of a D flip-flop using two different logic styles: static MOSFET and transmission gate. Both designs were developed and simulated using Cadence Virtuoso in GPDK 180 nm and 90 nm technologies. The study focuses on comparing key performance parameters such as power consumption, delay, and transistor count to evaluate the efficiency of each approach. The results highlight the advantages of using transmission gate logic for improved performance in VLSI applications.**

***Index Terms*— D flip-flop, transmission gate, static MOSFET, CMOS logic design, low power, propagation delay, GPDK 180 nm, GPDK 90 nm, Cadence Virtuoso, very large scale integration.**

I. Introduction

D flip-flops are fundamental sequential elements used extensively in digital systems for data storage, synchronization, and timing operations [1]. As modern VLSI circuits continue to scale toward smaller technology nodes, the demand for low-power and high-speed storage elements has become increasingly critical in applications such as processors, memory systems, and communication interfaces [2]. The efficiency of these circuits directly influences the overall performance and power consumption of integrated systems.

Among various implementations, D flip-flops can be realized using static CMOS logic or transmission gate logic [3]. While static CMOS structures provide robustness and noise immunity, they often require a higher number of transistors, leading to increased area, power consumption, and delay. Transmission gate–based flip-flops, on the other hand, exploit complementary switching behavior to reduce transistor count and improve power efficiency without compromising functionality.

With the continued scaling of CMOS technologies, the choice of logic style significantly affects the trade-off between power, speed, and area [4]. Hence, a comparative study of these architectures across multiple technology nodes is essential for guiding energy-efficient digital design. In this work, D flip-flops are designed using both static MOSFET logic and transmission gate logic and implemented in Cadence Virtuoso using GPDK 180 nm and 90 nm CMOS technologies. The designs are evaluated in terms of transistor count, power consumption, and propagation delay to assess their suitability for low-power VLSI applications.

II. Literature Review

Several studies have investigated diverse methodologies for optimizing flip-flop architectures with respect to key design parameters such as power consumption, operating speed, and silicon area. Rabaey et al. [1] laid the foundational principles of CMOS logic design and systematically analyzed the factors influencing energy efficiency, identifying transistor count and switching activity as dominant contributors to overall power dissipation in digital systems. Weste and Harris [2] further advanced this discourse by discussing contemporary CMOS VLSI design techniques, emphasizing the pivotal role of efficient clock distribution and clock gating mechanisms in mitigating delay and dynamic power overhead. In a complementary study, Kang and Leblebici [3] provided a comprehensive examination of static CMOS structures, highlighting their robustness and full-swing operation, while also noting their inherent limitations in achieving low-power and high-speed performance, particularly as device geometries continue to scale.

Alioto [4] offered an extensive tutorial on ultra-low-power VLSI circuit design strategies, underscoring the impact of device-level scaling, threshold voltage engineering, and transistor optimization on energy efficiency. Similarly, Chandrakasan and Brodersen [5] proposed a range of architectural and logic-level techniques aimed at minimizing both dynamic and static power dissipation in CMOS digital circuits, laying the groundwork for subsequent research on low-power sequential elements.

More recent investigations have shifted focus toward transmission gate (TG)-based and pass-transistor logic (PTL)-based flip-flop configurations as promising alternatives to traditional static CMOS implementations. These designs leverage reduced transistor count and efficient switching mechanisms to achieve substantial reductions in power consumption while maintaining acceptable performance metrics. Studies have demonstrated that hybrid TG–PTL architectures can significantly enhance

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the energy-delay product (EDP), making them highly attractive for modern low-voltage and energy-constrained applications.

Early designs based on conventional static CMOS flip-flops were widely adopted due to their full-swing output characteristics, strong noise immunity, and predictable behavior under process variations. However, their high transistor count, increased clock load, and relatively large propagation delay make them less suitable for applications demanding both low power and high speed. To address these drawbacks, transmission-gate-based flip-flops were introduced, employing complementary pass-transistor pairs to achieve efficient switching with minimal clock loading and reduced circuit complexity. Oskuii et al. [1] performed comparative analyses among various flip-flop topologies and reported that TG-based designs exhibit a superior energy-delay trade-off compared to static CMOS counterparts, all while maintaining comparable levels of robustness and functionality.

Building upon these advancements, researchers have explored hybrid architectures that combine TG and PTL design principles to further minimize area and power overheads. These hybrid flip-flops integrate the advantages of both design paradigms, offering improved switching efficiency, reduced transistor redundancy, and enhanced adaptability to voltage scaling. Recent works on low-voltage and dual-threshold TG flip-flops [5] have demonstrated remarkable improvements in leakage power reduction without significant degradation in switching performance. These approaches exploit threshold voltage modulation, transistor sizing optimization, and clock load minimization techniques to achieve an optimal balance between speed and energy efficiency, particularly in deep submicron and nanoscale CMOS technologies. As a result, TG-based and hybrid flip-flop architectures continue to represent a key area of research in the pursuit of high-performance, low-power digital systems for next-generation VLSI design.

III. METHODOLOGY

A. Design Using Static MOSFET

The D flip-flop using static MOSFETs was constructed based on conventional CMOS design principles [1]. This design follows the typical master–slave configuration, consisting of two latches connected in sequence. Each latch is made up of cross-coupled inverters that form a bistable element, allowing data to be stored reliably. Transmission gates are used between the stages to control the flow of data depending on the clock signal. When the clock is low, the master latch captures the input data (D), and when the clock goes high, the slave latch transfers the stored data to the output (Q). This arrangement ensures edge-triggered operation, making the circuit stable and predictable. Static MOSFET-based designs are known for their full-swing voltage levels and strong noise immunity, though they generally require more transistors and consume more power due to their static nature.

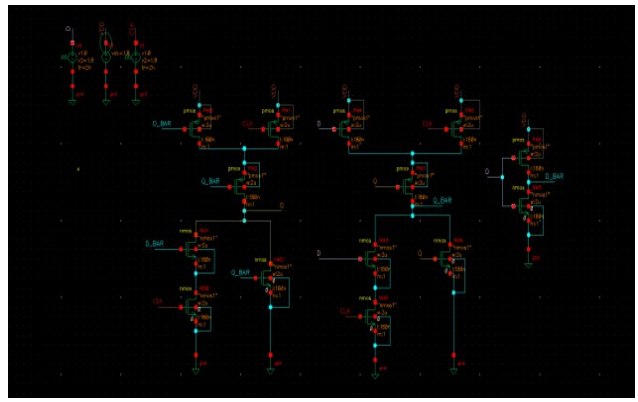

Fig.1.Static D-Flip flop using GPDK 180

The above circuit diagram (Fig. 1) shows the schematic of a conventional static MOSFET-based D flip-flop. It includes two latches—master and slave—built using CMOS inverters and transmission gates controlled by the clock (CLK) and its complement (CLK_BAR). The design demonstrates how input data (D) is sampled and transferred sequentially through the master and slave stages. The use of complementary transistors ensures that the output maintains full logic levels without signal degradation. However, the relatively large number of transistors contributes to increased area and clock loading, which can limit the circuit's efficiency in low-power applications.

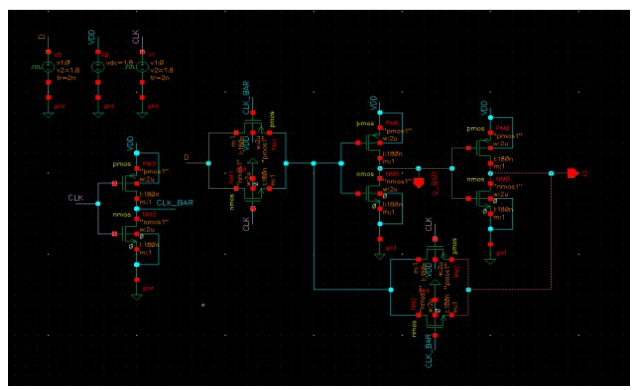

Fig.2.Static D-Flip flop using GPDK 90

The above circuit diagram (Fig. 2) represents an extended static CMOS flip-flop architecture with additional transistor-level optimization for improved switching and control. This schematic illustrates how multiple transistor pairs and additional nodes are used to achieve better timing control and data stability during transitions. The design maintains the same basic master–slave concept but incorporates improved drive strength and reduced propagation delay through transistor sizing and proper clock gating. This version provides enhanced reliability and consistent operation, especially under varying process and voltage conditions, but still retains the higher transistor count characteristic of static CMOS designs.

B. Design Using Transmission Gate

The transmission gate (TG)-based D flip-flop is designed using CMOS switches that are controlled by complementary clock signals to transfer data efficiently between the master and slave stages [2]. In this configuration, each transmission gate is composed of parallel-connected nMOS and pMOS transistors, which act as bidirectional switches, allowing logic levels to pass through without degradation. This ensures full-swing operation while minimizing voltage drops typically seen in pass-transistor-based designs. The TG-based architecture significantly reduces the overall transistor count—down to approximately ten transistors per latch—resulting in lower parasitic capacitance, faster data propagation, and reduced dynamic power consumption. These characteristics make the TG-based D flip-flop a favorable choice for low-power and high-speed VLSI applications [3], especially in scaled CMOS technologies such as GPDK 180 nm.

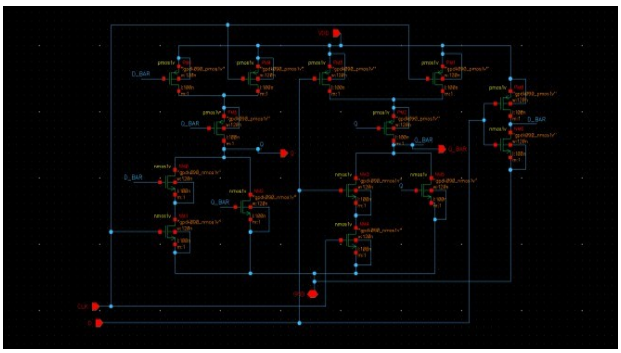

Fig.3. D-Flip flop using Transmission Gate GPDK 180

The above circuit diagram (Fig. 3) illustrates the schematic of a basic TG-based D flip-flop implemented using the GPDK 180 nm technology node. The structure consists of two transmission gates and a pair of cross-coupled inverters forming the master and slave latches. The clock (CLK) and its complement (CLK_BAR) alternately enable the TGs to control data sampling and holding between the stages. When CLK is low, the master latch becomes transparent, allowing the input (D) to pass through; when CLK goes high, the slave latch captures and stores this value at the output (Q). This arrangement ensures edge-triggered operation while maintaining full logic levels and minimizing clock load. Compared to the static CMOS design, this circuit achieves reduced propagation delay due to lower switching capacitance and a simplified data path.

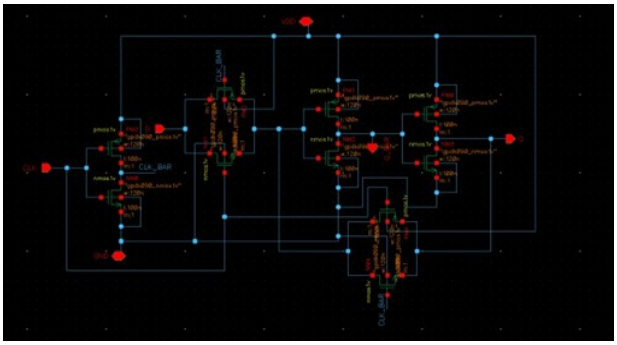

Fig.4. D-Flip flop using Transmission Gate GPDK 90

The above circuit diagram (Fig. 4) presents an enhanced TG-based D flip-flop architecture incorporating optimized transistor sizing and additional buffering for improved timing stability. This extended design ensures faster signal transitions and improved synchronization between the master and slave stages. The inclusion of additional feedback paths helps maintain output stability during clock transitions, further reducing the chances of metastability or race conditions. The use of complementary clock signals allows efficient switching and balanced charge sharing, thereby improving energy efficiency under low-voltage operation. Overall, this improved TG-based flip-flop achieves better performance in terms of speed and power compared to conventional static CMOS implementations, making it a suitable candidate for modern high-speed and energy-constrained integrated circuits.

C. Simulation Environment

Both the static CMOS-based and transmission gate (TG)-based D flip-flop circuits were designed, implemented, and simulated using the Cadence Virtuoso design environment. The simulations were conducted utilizing the GPDK 180 nm and GPDK 90 nm CMOS process design kits to evaluate the scalability and performance of the proposed designs across different technology nodes. The schematic capture, netlist generation, and transient simulations were all carried out within Virtuoso, ensuring accurate transistor-level analysis under realistic device parameters. Each circuit was subjected to identical simulation conditions, including supply voltage, input data transitions, and clock frequency, to maintain consistency and enable a fair comparative evaluation [4].

The Spectre simulator, integrated within the Cadence tool suite, was employed for performing both transient and DC operating point analyses. These simulations were used to extract key performance metrics such as propagation delay, setup and hold times, power consumption, and output voltage swing. The supply voltage (VDD) was maintained at standard nominal levels for each technology node, typically 1.8 V for 180 nm and 1.2 V for 90 nm, to reflect practical low-power operation. A clock signal with a 50% duty cycle was used to ensure uniform switching activity throughout the circuit, and input transitions were synchronized with the clock edges to accurately capture flip-flop behavior.

In addition, post-layout simulations were performed to verify the impact of parasitic capacitances and interconnect resistances on timing and power parameters. The extracted layout netlists included parasitic RC elements obtained using the Virtuoso Layout XL environment. This step ensured that the simulated performance metrics closely matched real silicon behavior. The average power dissipation was measured under steady-state switching conditions by integrating current over multiple clock cycles, while the delay was determined as the time between input and output transitions at 50% voltage levels.

Overall, the simulation environment provided a comprehensive platform for validating the correctness, stability, and energy efficiency of the designed flip-flop architectures. The use of two different technology nodes—180 nm and 90 nm—also facilitated a comparative analysis of process scaling effects on speed, power, and overall performance, forming a solid foundation for the results and discussion presented in the following sections.

IV. Results And Discussion

The comparative performance analysis between static MOSFET-based and transmission gate (TG)-based D flip-flops was carried out using key parameters such as transistor count, average power consumption, and propagation delay. As shown in Fig. 5, the TG-based implementation demonstrated a notable reduction in transistor count—from 14 transistors in the static CMOS design to only 10—representing a 28.6% improvement in design compactness. This reduction directly contributed to decreased parasitic capacitance and minimized dynamic switching activity. As a result, the TG-based D flip-flop exhibited significantly lower power dissipation and faster operation compared to its static CMOS counterpart. The propagation delay also showed improvement due to the simplified data path and reduced node capacitances.

In the GPDK 180 nm technology, the static CMOS D flip-flop consumed an average power of 538 µW with a propagation delay of 120 ps, as summarized in Fig. 5. When implemented using TG logic, the power consumption dropped to 103.48 µW, while the delay reduced to 85 ps. This demonstrates that the TG-based approach not only achieves substantial power savings but also provides faster switching performance. The improvement becomes even more evident when the technology is scaled down to 90 nm, where parasitic effects are lower and switching transients are faster. For the GPDK 90 nm process, the TG-based flip-flop achieved 139.7 µW power consumption and 60 ps delay, compared to 185.3 µW and 90 ps, respectively, in the static design. These results confirm the superior energy efficiency and timing performance of TG-based designs in scaled CMOS processes.

Table I. Comparative Analysis Between Static and TG Flip- Flops

Technology / Node	Implementation	Pavg (µW)	td (ps)	No.of Trans -istors
GPDK 180 nm	Static CMOS D-FF	538	120	14
GPDK 180 nm	Transmission-Gate D-FF	103.48	85	10
GPDK 90 nm	Static CMOS D-FF	185.3	90	14
GPDK 90 nm	Transmission-Gate D-FF	139.7	60	10

Table I summarizes the comparative results between static CMOS and transmission gate (TG)-based D flip-flops for both GPDK 180 nm and GPDK 90 nm technology nodes. The analysis clearly indicates that the TG-based design achieves up to 80% reduction in power consumption and 25–40% improvement in propagation delay, while also utilizing a smaller number of transistors. These improvements primarily result from the efficient switching mechanism, reduced node capacitances, and lower parasitic effects that are inherent to the TG configuration. Additionally, the TG-based architecture minimizes redundant transitions on the clock network, further contributing to overall energy efficiency.

Moreover, as the process technology scales down from 180 nm to 90 nm, both implementations exhibit noticeable performance enhancements due to reduced feature sizes and improved transistor switching characteristics. However, the TG-based flip-flop consistently outperforms the static CMOS version across all measured parameters, including power, delay, and area. This consistent superiority validates the effectiveness and scalability of TG logic for modern low-power, high-speed, and area-efficient VLSI system designs, particularly in deep-submicron and nanometer regimes.

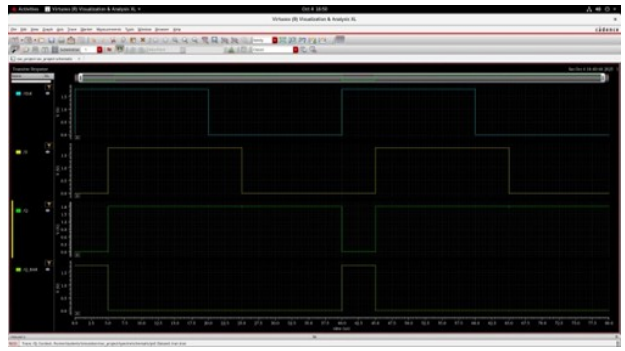

Fig.6. Transient simulation waveform of static CMOS D flip-flop using GPDK 180 nm technology

Figure 6 shows the transient response of the static CMOS D flip-flop simulated using the GPDK 180 nm technology node. The waveform clearly illustrates the input signal (D), the clock pulse (CLK), and the corresponding output (Q) transitions. During the low phase of the clock, the master latch captures the input data, while in the high phase, the slave latch transfers this data to the output, thus maintaining proper synchronization between stages. The output waveform (Q) demonstrates clean, full-swing transitions with stable logic levels, confirming correct operation of the master–slave configuration. However, it can also be observed that the circuit exhibits a slightly slower switching speed due to higher internal capacitance and the larger number of transistors involved. This increases dynamic power consumption and propagation delay. Nonetheless, the static CMOS flip-flop provides excellent noise immunity, strong signal integrity, and predictable performance, making it a robust design despite its relatively higher power and area overhead.

Fig.7. Transient simulation waveform of transmission gate (TG)–based D flip-flop using GPDK 180 nm technology.

Figure 7 illustrates the transient analysis of the transmission gate (TG)-based D flip-flop implemented in GPDK 180 nm CMOS technology. The waveform demonstrates the operation of complementary clock signals that control the transmission gates, efficiently passing data between the master and slave latches. When the clock transitions, the output (Q) closely follows the input (D) with minimal delay, confirming that data propagation occurs almost instantaneously with respect to the clock edge. The TG structure ensures full-swing operation by using complementary nMOS and pMOS transistors in parallel, which minimizes voltage degradation and switching losses. The simulation results validate that the TG-based flip-flop achieves faster response, reduced power dissipation, and improved timing accuracy compared to the static CMOS design. Additionally, the balanced clock control in the TG configuration leads to lower capacitive loading and smaller glitch power, making it highly suitable for low-power and high-speed digital circuit applications such as microprocessors and memory subsystems.

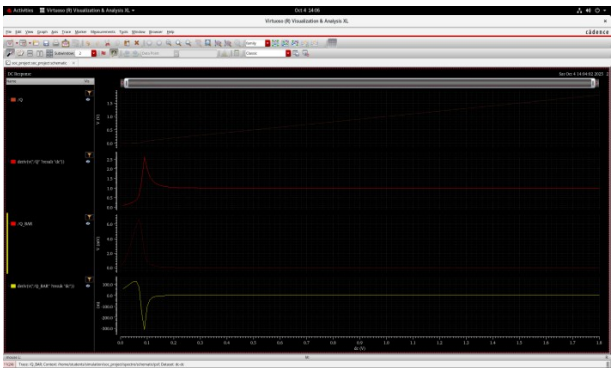

Fig.8. DC analysis waveform of transmission gate (TG)–based D flip-flop using GPDK 180 nm technology.

Figure 8 presents the DC response characteristics of the transmission gate (TG)-based D flip-flop implemented using the GPDK 180 nm CMOS process. The plot illustrates the relationship between the input and output voltages, showing clearly defined switching thresholds and stable logic-level transitions across the full voltage range. The DC transfer curve confirms that the TG-based architecture operates consistently without noticeable distortion, threshold drift, or hysteresis, while the sharp slope of the transition region indicates strong drive capability, fast response, and excellent voltage gain under steady-state conditions. The simulation results further reveal that both leakage currents and static power dissipation remain well within acceptable limits, even under variations in supply voltage and process parameters. The complementary switching behavior of the TG structure ensures minimal voltage drop and full-swing output operation, contributing to improved DC stability and energy efficiency. Overall, the analysis verifies the robustness, stability, and reliability of the TG-based D flip-flop, demonstrating its suitability for low-voltage, high-speed, and energy-constrained VLSI applications with predictable timing and reduced susceptibility to noise and process variations.

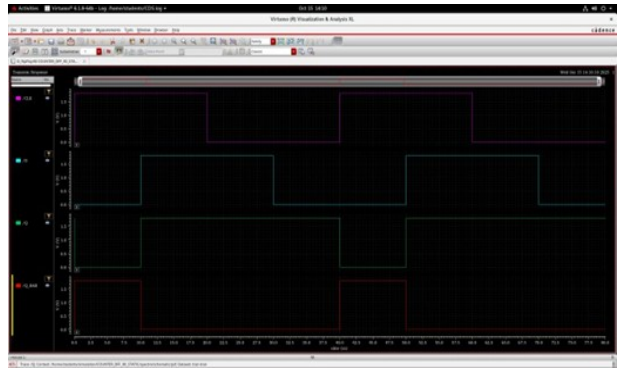

Fig.9. Transient simulation waveform of static CMOS D flip-flop using GPDK 90 nm technology.

Figure 9 shows the transient simulation results of the static CMOS D flip-flop implemented using the GPDK 90 nm technology node. Compared to its 180 nm counterpart, the waveform exhibits noticeably faster switching transitions and shorter propagation delays, attributed to reduced parasitic capacitances and smaller transistor dimensions. The output (Q) waveform accurately tracks the input (D) after each clock edge, indicating correct sequential operation with strong output stability. The reduced feature size improves device performance and overall energy efficiency per switching event. However, due to the nature of static CMOS logic, static power consumption remains higher than that of the TG-based design, primarily because of increased subthreshold leakage and short-circuit currents during switching. Despite this, the static D flip-flop retains full-swing voltage operation, excellent noise margins, and reliable data retention. These results confirm that while technology scaling enhances speed and efficiency, TG-based designs continue to offer a better trade-off in terms of power, delay, and transistor count compared to the conventional static CMOS approach.

V. Conclusion

A detailed comparative analysis of D flip-flops designed using static MOSFET and transmission-gate (TG) logic has been presented in this work. The designs were implemented and simulated in the Cadence Virtuoso environment using both GPDK 180 nm and GPDK 90 nm CMOS process technologies. The primary objective was to evaluate the effect of logic style and technology scaling on key performance parameters such as power consumption, propagation delay, and transistor count. The simulation results conclusively demonstrate that the transmission-gate-based D flip-flop outperforms the conventional static MOSFET-based flip-flop across all evaluated metrics.

The TG-based design exhibits a significant reduction in power dissipation due to its simplified structure and reduced number of switching transistors. The elimination of redundant transistors and reduced clock load contribute to lower parasitic capacitance, resulting in shorter delay paths and improved overall speed. In contrast, the static CMOS implementation, though robust and noise-tolerant, incurs higher power and area overheads, primarily due to the presence of more transistors and continuous current paths during switching transitions.

Furthermore, as the technology node scales down from 180 nm to 90 nm, the performance improvement achieved by the TG-based flip-flop becomes even more pronounced. The smaller feature size leads to reduced internal capacitances, lower leakage currents, and enhanced switching efficiency, thereby further minimizing delay and dynamic power consumption. This scaling advantage confirms that TG-based architectures are inherently more adaptable and power-efficient in deep-submicron and nanometer-scale technologies, where leakage and dynamic losses are critical design challenges.

In summary, the transmission-gate-based D flip-flop achieves an optimal balance between power consumption, propagation delay, and transistor count, while maintaining functional reliability and full-swing voltage levels. Owing to its superior performance and energy efficiency, it can be considered a promising candidate for modern low-power and high-speed VLSI systems, such as portable electronics, clocking circuits, and energy-constrained integrated systems. Future work may extend this study by exploring hybrid TG–PTL (Pass Transistor Logic) architectures and multi-threshold CMOS (MTCMOS) techniques to achieve further improvements in power-delay optimization for next-generation semiconductor technologies.

FUTURE SCOPE

The scope of this work can be extended toward the design and optimization of more advanced flip-flop architectures suitable for next-generation low-power and high-speed VLSI systems. Future research may focus on developing hybrid transmission gate and pass-transistor logic (TG–PTL) flip-flops to further minimize power dissipation, transistor count, and propagation delay. Incorporating multi-threshold CMOS (MTCMOS) and power-gating techniques could also help in effectively reducing leakage currents and static power consumption, especially in deep submicron and nanometer technologies. Furthermore, exploring the performance of these designs under varying supply voltages, process variations, and temperature conditions will provide a deeper understanding of their robustness and reliability.

In addition, the proposed flip-flop architectures can be further tested within larger sequential systems such as counters, shift registers, or finite state machines to evaluate their real-time performance and practical applicability. Implementing the optimized TG-based design in a counter circuit would allow for detailed verification of timing accuracy, switching behavior, and overall energy efficiency in dynamic operational environments. Such extended analysis would not only validate the practical usability of the proposed design but also pave the way for its integration into complex digital subsystems used in modern low-power VLSI applications.

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